

Module 5: Action — Movement Is Prediction in Motion

Learning Objectives

1. Understand that every physical movement is your brain **predicting the future position of your body** and sending commands to get there.
2. Learn why **practice works** at a neurological level and how your brain builds automatic movement patterns.
3. Discover how your body takes **invisible actions** (immune responses, digestion, hormone release) using the same predict-and-adjust loop.

Introduction

Think about catching a ball. Sounds simple, right? But watch what actually happens: Your eyes track the ball. Your brain calculates its speed, trajectory, and spin. It predicts where the ball will be in about half a second. It sends signals to your legs (move right), your arm (extend), your hand (open, then close at exactly the right moment). All of this happens before the ball arrives. You're not reacting to where the ball is — you're acting on where your brain **predicts** it will be.

If your brain waited until the ball was in your hand to start moving, you'd never catch anything. Movement is prediction in motion. Every step you take, every pencil you pick up, every bite of food you guide to your mouth — your brain is running ahead of the present moment, predicting and executing.

Key Concepts

1. Your Brain Moves You Before You Know It

Most of your movements happen faster than conscious thought. When you trip, your arms fly out to catch yourself before you even realize you're falling. When something flies toward your face, you flinch before you can identify what it is.

These fast movements are powered by **motor predictions** — pre-programmed responses your brain has ready to go. Your brain predicts "if I'm falling, I need to extend my arms" and executes the plan without waiting for your conscious mind to weigh in. This is faster and safer than thinking it through.

Even "voluntary" movements are mostly predicted. When you decide to pick up your water bottle, your brain doesn't calculate every muscle contraction from scratch. It pulls up a pre-built prediction model of "how to reach and grasp" and adjusts it slightly for the specific bottle position. That's why familiar movements feel effortless — your brain has excellent predictions for them.

2. Why Practice Actually Works

When coaches say "practice makes perfect," here's what's actually happening in your brain.

The first time you try a new movement (a guitar chord, a soccer kick, a dance move), your brain's prediction model is rough and inaccurate. You send motor commands, observe the results, notice the prediction errors, and update the model. Your fingers land on the wrong frets. Your kick goes wide. Your timing is off.

Each repetition refines the model. Your brain adjusts the predictions: "a little more wrist rotation," "shift weight to the left foot earlier," "start the arm swing a fraction of a second sooner." After hundreds or thousands of repetitions, the prediction model becomes incredibly precise. The movement feels automatic because your brain's predictions are now so accurate that almost no correction is needed.

This is also why you can't just watch someone do something and immediately copy it. Watching builds a visual model, but your motor system needs its own prediction errors to build its own model. You have to *do* it, mess up, and let your brain update.

3. Your Body Acts Without Asking Permission

Not all actions require muscles you control. Your body takes critical actions every moment through systems you can't consciously command.

Your immune system acts by deploying white blood cells to infection sites. Your digestive system acts by releasing enzymes to break down food. Your endocrine system acts by pumping hormones into your bloodstream. All of these are actions in the active inference sense — your body predicts what's needed and responds.

When you eat, your body predicts the nutrient content of the food based on past experience and prepares the appropriate digestive enzymes before the food even reaches your stomach. If you eat something unexpected (much fattier or much spicier than predicted), your body scrambles to adjust — which is why unfamiliar foods can sometimes cause stomach upset. Your digestive system had the wrong prediction.

Try It!

Motor Learning Experiment. Try bouncing a ball off a wall and catching it with your non-dominant hand. Do it 10 times and rate your performance from 1-10. Then do 10 more tries every day for five days (50 total reps). Rate yourself after each set. You should see your performance improve as your brain builds a better motor prediction model. Bonus: if you switch to a differently-sized or differently-weighted ball, notice how your performance drops temporarily as your brain adjusts its predictions.

Summary

Every movement is your brain **predicting** where your body needs to be and sending commands to get there. **Practice** works because each repetition generates prediction errors that refine your brain's motor models, making movements smoother and more automatic. Your body also takes **invisible actions** — immune responses, digestion, hormone release — using

the same predict-and-adjust loop. Next: **Learning** — how your body rewires itself based on all these experiences.